

ST MARY RESERVOIR, AB, Canada

WMO: 718370

Lat: 49.3614N

Lon: 113.1100W

Elev: 3701

StdP: 12.83

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.9	-11.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	86.2	N/A	82.6	N/A	79.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-25.9	90.9	4.5	4.3	-29.2	94.0	-31.8	96.5	-34.4	98.9	-37.7	102.0
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	23.1	20.0	30.0	40.2	50.2	58.0	64.5	63.7	55.3	43.1	31.7	20.5		
	DBStd	19.21	14.09	15.35	12.93	9.19	7.87	6.05	5.67	6.00	8.92	9.90	12.86	14.50		
	HDD50	4481	834	841	621	311	97	8	0	2	48	252	554	913		
	HDD65	8609	1299	1261	1086	745	460	222	78	97	303	680	1000	1378		
	CDD50	1494	0	0	0	16	103	247	451	426	208	38	5	0		
	CDD65	144	0	0	0	0	1	11	63	56	13	0	0	0		
	CDH74	2260	0	0	0	7	52	168	893	855	280	5	0	0		
	CDH80	620	0	0	0	0	1	26	272	261	60	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	21.6	1.1	0.9	1.3	1.9	3.0	4.2	1.9	1.8	1.9	1.3	1.2	1.1		
	PrecMax	33.2	1.7	1.9	2.5	3.7	6.5	12.7	6.8	4.6	5.3	3.1	2.4	2.3		
	PrecMin	13.5	0.3	0.2	0.5	0.7	0.4	0.9	0.1	0.1	0.2	0.2	0.4	0.4		
	PrecStd	4.7	0.4	0.4	0.5	0.7	1.5	2.6	1.5	1.1	1.1	0.7	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	51.7	56.5	74.1	78.9	84.1	91.4	90.7	85.7	72.9	63.4	50.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	45.6	45.4	53.0	65.1	75.0	78.4	86.8	86.2	81.3	66.6	55.8	45.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	42.6	41.8	49.3	59.6	70.7	75.1	82.5	82.5	76.9	61.6	51.1	41.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	40.1	38.8	45.3	54.7	66.0	71.5	78.9	78.9	71.5	56.8	47.2	38.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	17.1	17.7	19.3	20.5	20.4	24.1	25.1	22.4	18.8	15.9	15.5		
		MCDBR	16.3	17.8	21.6	27.3	28.6	26.4	30.1	31.8	31.3	24.5	20.1	16.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.256	0.277	0.318	0.328	0.324	0.329	0.345	0.309	0.277	0.253	0.229		
	taud		2.429	2.446	2.480	2.383	2.415	2.470	2.461	2.395	2.473	2.542	2.501	2.437		
	Ebn at Noon		267	290	301	295	295	296	293	283	285	279	262	259		
	Edh at Noon		19	24	28	35	35	33	33	34	28	22	18	16		
(44) All-Sky Solar Radiation	RadAvg		394	712	1105	1528	1820	2034	2253	1794	1275	792	442	315		
	RadStd		33	53	96	101	158	154	121	111	104	78	34	30		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page